

1 Notes

- A shortest may not exist when:
 - Negative weight cycle
 - The graph is not connected
- Two Properties of The Shortest Path
 - The optimal substructure of dynamic shortest path: A subpath of a shortest path itself is a shortest path
 - Triangle inequality
 - * For all $u, v, x \in V$ we have $\delta(u, v) \leq \delta(u, x) + \delta(x, v)$
 - * Shortest path from u to v is at most any particular path, e.g., the blue chain.
- Tractable Problems
 - Problems solvable in a polynomial time
- $P \subset NP$
- NPC: If any other problem (i.e., NP-Hard problems) p in NP is polynomial reducible to problem q

2 Psuedocode

2.1 Dijkstra(G,s)

```
d[s] = 0
for each vertex v in V-{s}
    d[v] = infinity
    S = emptySet
    Q = V

while Q is not empty
    u = Extract_Min(Q)
    S = S ∪ {u}
    for each vertex v adjacent to u
        if d[v] > d[u] + w(u,v)
            d[v] = d[u] + w(u,v)
```

2.2 Bellman-Ford

Given $G=(V,E,w)$ and source vertex s
 $d[x]$ = distance estimate from s to x

```
Bellman-Ford( $G,s$ )
 $d[s] = 0$ 
for each vertex  $v$  in  $V-\{s\}$ 
     $d[v] = \text{infinity}$ 

for  $i=1$  to  $|V|-1$ 
    for each edge  $(u,v)$  in  $E$ 
        if  $d[v] > d[u] + w(u,v)$ 
             $d[v] = d[u] + w(u, v)$ 

for each edge  $(u, v)$  in  $E$ 
    if  $d[v] > d[u] + w(u, v)$ 
        report that a negative-weight cycle exists
```

2.3 Floyd-Warshall

```
 $D^{\wedge}(0) = W$ 
for  $k = 1$  to  $n$  do
    for  $i = 1$  to  $n$  do
        for  $j = 1$  to  $n$  do
             $d_{ij}(k) = \min\{d_{ij}^{(k-1)}, d_{ik}^{(k-1)} + d_{kj}^{(k-1)}\};$ 

Return  $D^{\wedge}(n)$ 
```

2.4 Ford Fulkerson

```
FORD-FULKERSON( $G, s, t$ )
    for each edge  $(u,v)$  in  $E(G)$ 
        do  $f[u, v] = 0$ 
            $f[v, u] = 0$ 
    while there is a path  $p$  from  $s$  to  $t$  in the residual network  $G_f$ 
         $cf(p) = \min\{cf(u, v)-f[u, v]: (u, v) \text{ is in } p\}$ 
        for each edge  $(u, v)$  on  $p$ 
            if  $(u, v)$  in  $E$ 
                 $f[u, v] = f[u, v] + cf(p)$ 
            else  $f[v, u] = f[v, u] - cf(p)$ 
```

3 Complexities

Dijkstra's $\Theta(V) \times T_{\text{Extract-Min}} + \Theta(E) \times T_{\text{Decrease-Key}}$

Bellman-Ford $\mathcal{O}(VE)$

Floyd Warshall $\mathcal{O}(V^3)$

Ford-Fulkerson $\mathcal{O}(E|f^*|)$

Edmond Carp $\mathcal{O}(VE^2) = \mathcal{O}(V^5)$